

CHAPTER 1:- INTRODUCTION

1 Micro, Macro, and Engineering Economics

A. Microeconomics

- **Definition:** Study of economic behavior of **individual units**—consumers, firms, industries.
- **Focus Areas:**
 - Demand & supply of goods
 - Price determination
 - Production and cost analysis
 - Market structures (perfect competition, monopoly)
- **Examples:**
 - A company deciding how many units of a product to produce.
 - Consumer choosing between two brands.

B. Macroeconomics

- **Definition:** Study of the economy as a **whole**.
- **Focus Areas:**
 - National income
 - Inflation
 - Unemployment
 - Monetary and fiscal policy
 - Economic growth
- **Examples:**
 - Government deciding tax rates.
 - Central bank controlling money supply.

C. Engineering Economics

- **Definition:** Application of economic principles to **engineering problems** for optimal decision-making.
- **History & Evolution:**
 - Emerged in early 20th century with industrialization.
 - Developed to help engineers compare investment alternatives.
- **Fundamental Principles:**

- **Applications:**
 - Selecting the best machine or technology
 - Designing cost-efficient structures and systems
 - Project feasibility analysis
 - Cost–benefit evaluation
 - Replacement and maintenance analysis
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1.2 Terminology Related to Engineering Economics

Key Terms

1. **Cost:** Expenditure required to produce a good or service.
Example: Cost of materials, labor.
 2. **Fixed Cost:** Does NOT vary with production level.
Example: Rent, salaries.
 3. **Variable Cost:** Changes with production.
Example: Raw materials.
 4. **Revenue:** Income generated from selling goods/services.
 5. **Depreciation:** Reduction in the value of assets over time.
 6. **Interest:** Cost of borrowing money or return on investment.
 7. **Present Value (PV):** Current worth of future cash.
 8. **Future Value (FV):** Value of cash at a future date.
 9. **Life-cycle Cost:** Total cost of a project from start to disposal.
 10. **Salvage Value:** Estimated value of an asset at the end of its life.
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1.3 Economic Decision and Role of Engineers in Decision Making

A. Economic Decision

Economic decisions involve comparing alternatives based on **cost, benefit, efficiency, and long-term economic impact**.

Examples:

- Choosing between two construction materials.
- Selecting energy-efficient machinery.
- Evaluating whether to repair or replace equipment.

Steps in Economic Decision Making

1. Define the problem.
2. Identify feasible alternatives.
3. Evaluate cash flows.
4. Consider constraints (budget, technology).
5. Apply economic methods (PW, FW, IRR, B/C ratio).
6. Select the best alternative.
7. Review and monitor performance.

B. Role of Engineers

Engineers play a key role because:

- They understand **technical feasibility**.
- They estimate **costs, efficiency, and performance**.
- They perform **economic analysis** of projects.
- They optimize resource usage.
- They ensure **safe, economical, and sustainable** implementation.

In summary: Engineers bridge the gap between **technical design** and **economic viability**.

1.4 Cash Flow and Cash Flow Diagram

A. Cash Flow

- Cash flow refers to **movement of money** in a project.
- Two types:
 - **Cash Inflow:** Revenue, savings, salvage value.
 - **Cash Outflow:** Initial investment, operating cost, maintenance.

Components of Cash Flow

- Initial cost (negative)
 - Operating cost
 - Maintenance/repair cost
 - Salvage value
 - Annual revenue/savings
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B. Cash Flow Diagram

A cash flow diagram is a **graphical representation** of inflows and outflows along a time axis.

Elements:

- **Horizontal axis:** Time (years)
- **Vertical arrows:**
 - Upward arrows represent **inflows (+)**
 - Downward arrows represent **outflows (-)**
- Interest rate is usually shown.

Example Cash Flow Diagram

Year: 0 1 2 3 4

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Outflow:

↓10000

Inflow: ↑3000 ↑3000 ↑3000 ↑3000

This diagram shows:

- Initial investment = Rs. 10,000 at Year 0
- Annual inflow = Rs. 3,000 for Years 1–4

CHAPTER 2:- MARKET ECONOMICS

2.1 Market, Demand, Supply, and Their Relationship

A. Market

- **Definition:** A platform where buyers and sellers interact to exchange goods and services.
- **Types of Markets:**
 - Perfect competition
 - Monopoly
 - Oligopoly
 - Monopolistic competition

B. Demand

- **Definition:** Quantity of a good or service a consumer is willing and able to buy at a given price.
- **Law of Demand:** Price $\uparrow \rightarrow$ Demand $\downarrow$ (inverse relationship)
- **Factors Affecting Demand:**
 - Income levels
 - Prices of related goods (substitutes/complements)
 - Consumer preferences
 - Government policies

C. Supply

- **Definition:** Quantity of a good or service producers are willing and able to sell at a given price.
- **Law of Supply:** Price $\uparrow \rightarrow$ Supply $\uparrow$ (direct relationship)
- **Factors Affecting Supply:**
 - Production cost
 - Technology
 - Taxes and subsidies
 - Price of inputs

D. Demand-Supply Relationship

- **Equilibrium Price:** Price at which quantity demanded = quantity supplied.

- **Excess Demand:** Price below equilibrium → shortage.
 - **Excess Supply:** Price above equilibrium → surplus.
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2.2 Elasticity, Application of Elasticity, and Government Policies

A. Elasticity

- **Price Elasticity of Demand (PED):** Measures sensitivity of quantity demanded to price changes.

$$PED = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in price}}$$

- **Elastic:** PED > 1 (highly responsive)
- **Inelastic:** PED < 1 (less responsive)
- **Unitary:** PED = 1
- **Income Elasticity of Demand:** Measures response of demand to changes in income.
- **Cross Elasticity of Demand:** Measures response of demand for one good to price change of another good.

B. Applications of Elasticity

- Pricing strategies for firms
- Tax incidence analysis
- Forecasting demand changes due to policy
- Decision on product substitutions

C. Government Policies

- **Price Controls:**
 - Price ceiling → prevents price above maximum (e.g., rent control)
 - Price floor → prevents price below minimum (e.g., minimum wage)
 - **Taxes & Subsidies:** Influence supply and demand
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2.3 Externality and Market Inefficiency

A. Externality

- **Definition:** Effect of a transaction on a third party not involved in the market.
- **Types:**
 - **Positive Externality:** Benefits others (e.g., vaccination, education)

- **Negative Externality:** Harms others (e.g., pollution, noise)

B. Market Inefficiency

- Occurs when resources are **not allocated optimally**.
- Causes include:
 - Externalities
 - Public goods
 - Information asymmetry

Example:

- Factory emits pollution → cost borne by society, not reflected in price → overproduction → market inefficiency
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2.4 Market Failure and Firm Behavior

A. Market Failure

- **Definition:** When markets fail to produce **efficient allocation of resources**.
- **Causes:**
 - Monopoly power
 - Public goods
 - Externalities
 - Information gaps

B. Firm Behavior

- **Objectives of Firms:**
 - Profit maximization
 - Cost minimization
 - Market share growth
- **Production Decisions:**
 - How much to produce
 - Price to charge
 - Choice of technology and inputs
- **Types of Firms:**
 - Perfectly competitive firms → price takers
 - Monopolistic firms → price makers

Key Note: Understanding market behavior helps engineers anticipate **economic consequences of projects**, set **prices**, and design cost-effective solutions.